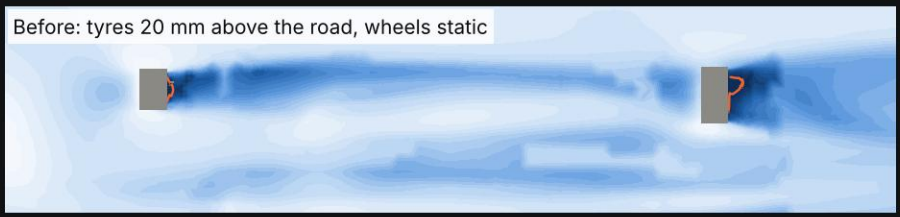
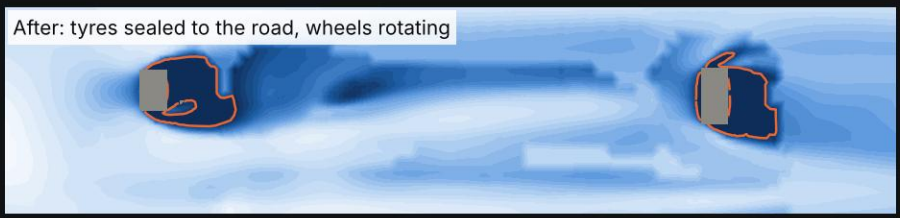
Closing the gap under the tyres

Contact patch and rotating wheels on a generic 2026 F1 model





Air speed 30 mm above the road, left half of the car, seen from above (dark = slow). 20 m/s.

Sealing the gap and spinning the wheels: drag +2.6%, downforce -13%. The air that used to run under the tyres now goes around them, and the wheel wakes grow.

Drag is grid-independent to under 1%. Downforce is not (slide 9).

Three ways to put a tyre on the road

Same car, mesh and settings for all three. Only the tyre–road contact and the wheel motion change.

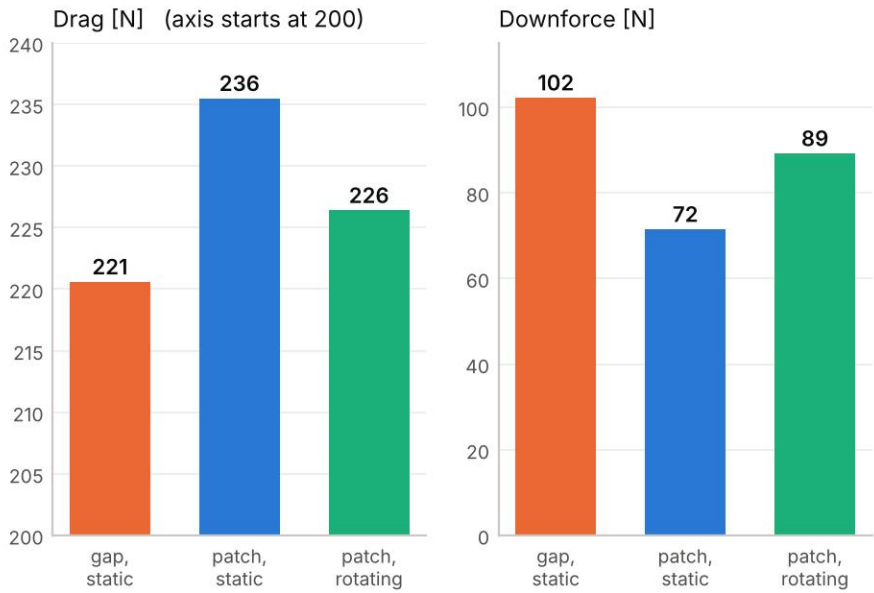


1 Gap, static wheels **2 Contact patch, static** **3 Contact patch, rotating**

Contact patch	120 mm block under each tyre, tread-wide, moving with the road
Rotating wheels	Tyre surface spins at road speed (rigid rotation)
Solver	simpleFoam, k-ω SST, Spalding wall function
Mesh	3 layers on the car, 1.43M cells (medium grid)
Flow	20, 50 and 80 m/s; 1% inlet turbulence
Checks	Momentum balance at 5 wake planes, all 14 runs

What it does to the forces

20 m/s, medium grid. Mean of iterations 2000–3000.



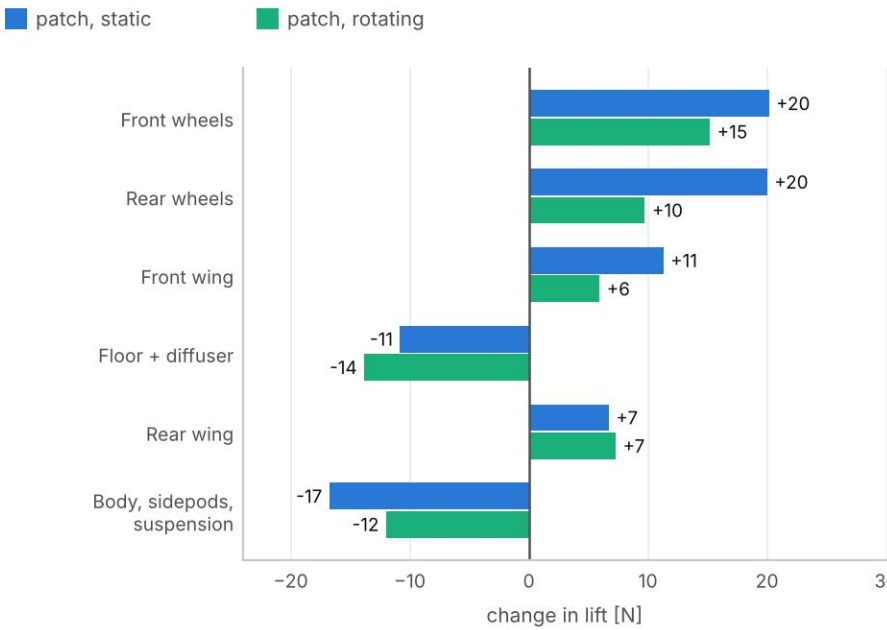
Sealing the gap with static wheels: drag +6.8%, downforce -30%. Then spinning the wheels gives part of it back: drag -3.9%, downforce +25%.

Net, the realistic model has +2.6% drag and -13% downforce vs the gap model, and the front axle carries 92% of it (was 85%).

Where the downforce went

Pressure lift by region, change from the gap model.

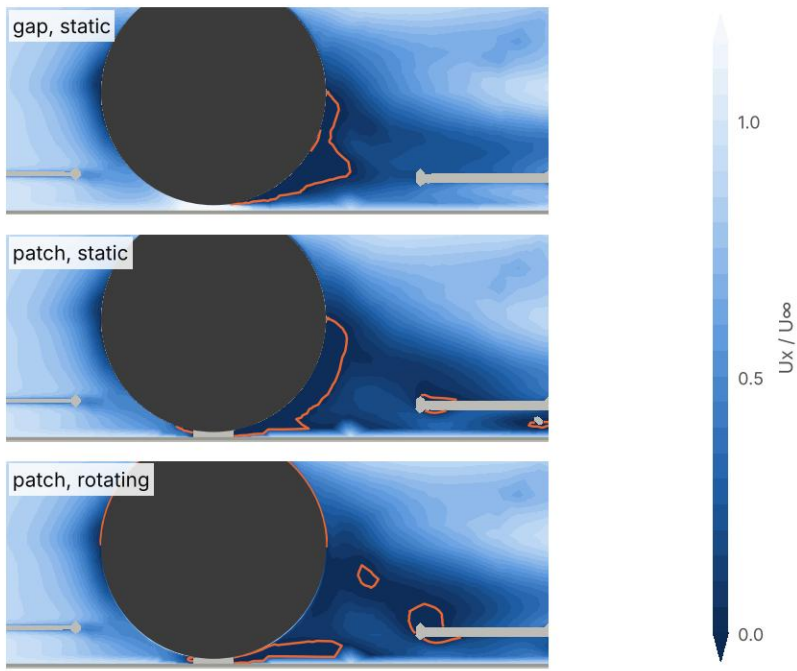
Positive = less downforce.



The tyres themselves lose the most: once sealed to the road, the air can no longer run underneath and suck them down (front +23 → +43 N, rear +30 → +50 N, static). The floor and body gain a little. Front-wheel drag: 40 → 53 N, the largest drag change on the car.

The front tyre, from the side

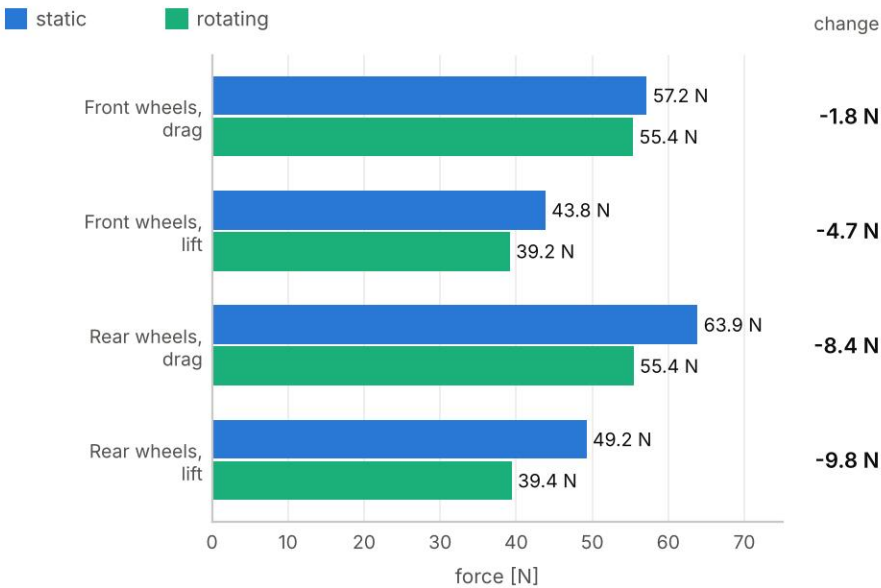
Streamwise air speed through the front-left tyre centre, 20 m/s. Orange line = air flowing backwards.



Gap: air jets through the 20 mm slot under the tyre (light streak at the road). Sealed: the jet is gone and the backflow region behind the tyre grows. Rotating: it shrinks again and hugs the road.

What spinning the wheels does

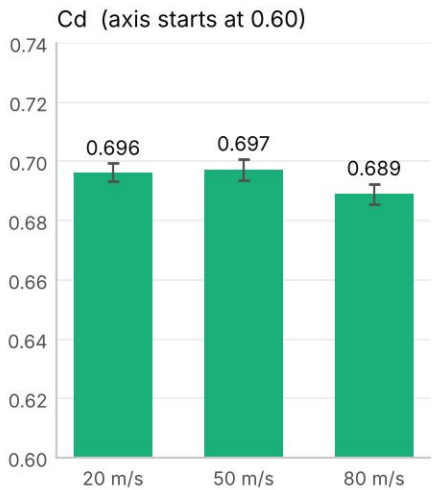
Wheel forces, contact-patch model, 20 m/s.
Static vs rotating wheels, everything else identical.

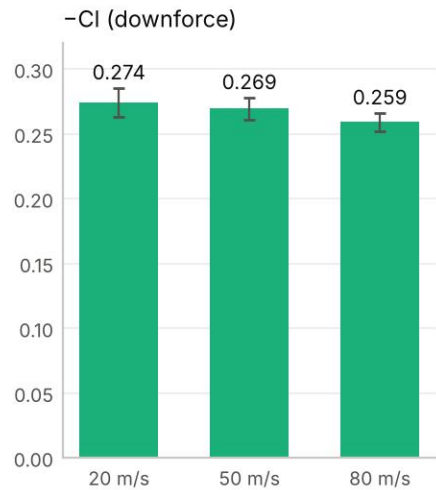


Spinning lowers both lift and drag on the wheels, as in classic wind-tunnel tests on isolated wheels (Fackrell & Harvey, 1975). The rear tyres change most: drag -13%, lift -20%. With the wheels unloaded, the car as a whole gets 25% more downforce than the static patch model.

20, 50 and 80 m/s

Realistic model (contact patch, rotating wheels). If the flow doesn't change character with speed, the coefficients shouldn't either.



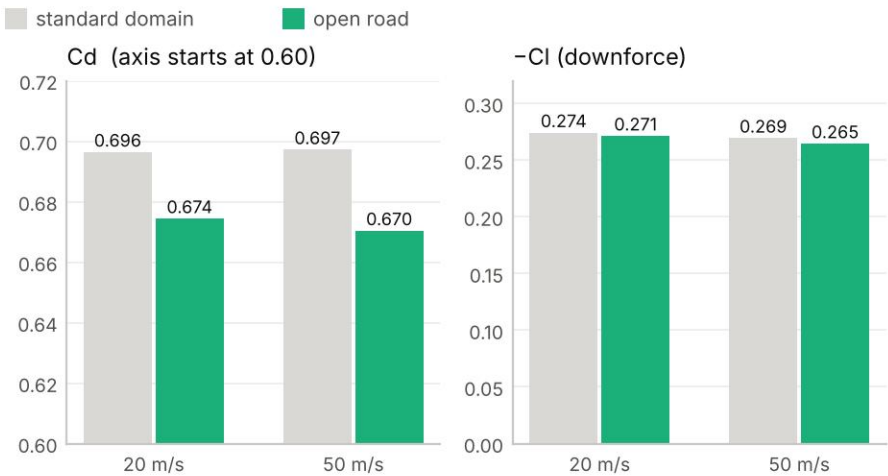


20 → 80 m/s: Cd -1.0%, downforce coefficient -5.5%.
Error bars: iteration-to-iteration scatter (1 std).

At 80 m/s, 32% of the car's surface sits above y+ 300 (20 m/s: 5%). The wall function covers it, but the 80 m/s downforce is the least certain number here.

Tunnel walls vs open road

Same car mesh in a much bigger domain (blockage 2.8% → 0.9%). Realistic model.

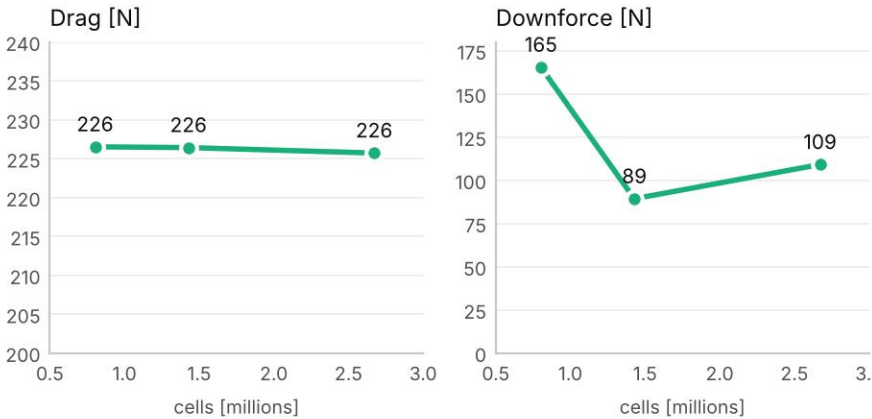


The tunnel walls squeeze the flow and add 3.2% (3.8% at 50 m/s) to the drag. Downforce moves by 1–2%, inside the run-to-run scatter. Every comparison in this post uses the same domain, so the changes between models hold either way.

The honest slide: mesh independence

Realistic model at 20 m/s on three grids.

0.80M / 1.43M / 2.67M cells.



Drag is grid-independent: Cd 0.694 ± 0.8% (GCI, Celik 2008).

Downforce is not. It zigzags between grids, so the standard GCI doesn't apply. Half the spread gives Cl -0.34 ± 0.12 (±35%).
By region: front wing -141 / -118 / -108 N, settling steadily; rear wing -85 / -57 / -72 N, oscillating. The finer grid also resolves slightly more of the car's surface (+4% area, coarse → fine).

So: trust the drag numbers. Treat downforce as a trend.

Other things I checked

Each one could have explained the results away.

1 Why is downforce 41% lower than in Post 2?

A control run with Post 2's wall function on the new mesh: 92% of the change comes from the new 3-layer near-wall mesh, 8% from the wall function. Mostly on the tyres in the 20 mm gap and the front wing.

2 Does the contact-patch size matter?

100 / 120 / 140 mm blocks: Cd 0.699 / 0.696 / 0.690, −Cl 0.282 / 0.274 / 0.281. Within the scatter.

3 Is the rotation set up right?

Every tyre face checked against rigid-body rotation before the run: worst error 0.27% of road speed.

4 Do the forces add up?

Body + wheels add up to the total on every run. Momentum balance at five wake planes matches the surface drag within ±1.4% (13 runs) and ±2.9% on the coarse grid.

5 Is a static wheel on a moving road real?

No. It's kept as a modelling step to separate the contact-patch effect from the rotation effect.

Three posts, one lesson

The F1 series in one slide.

Post 1 Mesh study

A 41% downforce difference lived in a 20 mm gap under the rear tyres.

Post 2 Wake drag

Drag is ~95% profile loss; check the wake, not just the forces.

Post 3 Contact patch

Sealing the gap and spinning the wheels: drag +2.6%, downforce -13%. Drag is grid-independent; downforce isn't yet.

The lesson: the hardest part of race-car CFD is where the car touches the road. Model it, then prove your mesh, before trusting a downforce number.

Next for a real downforce answer: a finer, more uniform mesh around the floor and rear wing, and a transient (DES) run.